

Symbolic Decarbonization and Stock Price Crash Risk: The Corporate Information Environment in the Dual-Carbon Era

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ABSTRACT

This paper examines whether symbolic decarbonization—the gap between firms’ green rhetoric and substantive environmental actions—predicts future stock price crash risk. Using a two-stage residual approach that isolates pure greenwashing from legitimate climate disclosure, we find that the greenwashing–crash association is sharply concentrated in state-owned enterprises and industrial firms—precisely where political incentives for green signaling are strongest. Two proposed transmission channels are examined but neither functions as a conventional mediator: greenwashing firms exhibit *lower* discretionary accruals (consistent with textual-for-financial substitution), yet $|DA|$ does not predict crash risk; greenwashing temporarily boosts short-term operating performance, and declining ΔROA strongly predicts elevated crash risk, but the mediating pathway through ΔROA is suppressive rather than transmissive. That neither channel mediates the relationship is a central finding—the true transmission mechanism remains unidentified. The greenwashing–crash association is further concentrated in firms without substantive green capital expenditure, consistent with a risk-buffer interpretation. Our findings establish symbolic decarbonization as a material, institutionally concentrated source of tail risk operating through channels distinct from financial opacity.

1. Introduction

In September 2020, China announced at the United Nations General Assembly its “dual-carbon” commitment: peak carbon emissions by 2030 and achieve carbon neutrality by 2060. This landmark policy pledge has fundamentally reshaped the institutional landscape facing Chinese listed firms. Regulators have issued a cascade of implementing directives—collectively termed the “1+N” policy framework—mandating enhanced environmental disclosure, carbon emission reporting, and green transition planning across all major industries. For firms, the dual-carbon era presents both an opportunity and a pressure: those perceived as environmentally responsible may benefit from lower financing costs, regulatory goodwill, and reputational premiums (Hong, Karolyi and Scheinkman, 2020; Bolton and Kacperczyk, 2021), while those perceived as laggards face escalating regulatory scrutiny and market penalties. This institutional shock has created powerful incentives for firms to *signal* environmental commitment—but not necessarily to deliver it.

The tension between green signaling and green substance gives rise to *symbolic decarbonization*: the managerial practice of saturating corporate disclosures with low-carbon rhetoric while undertaking little commensurate real environmental investment. Unlike financial misreporting, which leaves a verifiable audit trail, symbolic decarbonization operates in the relatively unregulated space of narrative disclosure—the Management Discussion & Analysis (MD&A) section of annual reports, where managers enjoy substantial discretion over tone, emphasis, and vocabulary (Loughran and McDonald, 2014). When external verification of environmental claims is costly and non-financial disclosure auditing remains less rigorous than financial statement auditing, firms can capture the valuation benefits of green signaling without incurring the costs of substantive decarbonization (Marquis, Toffel and Zhou, 2016). This moral hazard problem is particularly acute in China, where the regulatory framework governing MD&A textual disclosure is less prescriptive than its US counterpart and where state ownership introduces additional political incentives for environmental rhetoric (Piotroski, Wong and Zhang, 2015).

We argue that symbolic decarbonization is not merely a disclosure artifact—it is a material source of tail risk. The logic follows directly from the bad news hoarding theory of Jin and Myers (2006): when managers systematically conceal adverse information, the accumulated negative information is released suddenly when it exceeds a critical threshold, producing a stock price crash. Hutton, Marcus and Tehranian (2009) provide the canonical empirical bridge, showing that firms with higher discretionary accruals—a measure of financial opacity—exhibit significantly elevated future crash risk. We extend this logic from the financial to the environmental domain: greenwashing injects dense, ambiguous environmental narratives into corporate communications, impairing the ability of external stakeholders to monitor firm performance and facilitating the accumulation of hidden bad news. At the same time, resources diverted from long-gestation green capital projects to low-cost symbolic signaling generate a growing gap between market expectations and operational reality; when this gap becomes apparent, the valuation correction is sharp—a crash.

Prior research has explored the consequences of greenwashing through several lenses. Zhang, Kong and Hu (2024) document a positive association between ESG disclosure–performance gaps and crash risk in China. Dong and Ma (2025) find that excessive environmental information disclosure in CSR reports exacerbates crash risk through increased information opacity. However, these studies face a common measurement challenge: ESG rating-based greenwashing proxies are confounded by the well-documented divergence across rating agencies, which is driven primarily by measurement differences (56%) and scope differences (38%) rather than by genuine disagreement about firms’ underlying environmental performance (Berg, Kölbel and Rigobon, 2022). Moreover, existing studies do not distinguish between legitimate climate risk communication—firms operating in regions exposed to extreme weather, floods, or other physical climate hazards have valid reasons to discuss climate adaptation—and opportunistic greenwashing. Finally, prior work examines a single transmission channel (information opacity), leaving unexplored the expectation inflation mechanism: the possibility that green signaling generates temporary performance improvements whose subsequent reversal contributes to crash formation.

In this paper, we address these challenges with a two-stage residual approach. In the first stage, we decompose annual report green-word intensity (*WordTransition*) into: the portion explained by substantive green capital expenditures (*GreenCAPEX*), measured from project-level Construction-in-Progress accounting data matched against a green-technology keyword lexicon; the portion explained by objective physical climate risk exposure (*PhysicalRisk*), capturing legitimate risk communication; and an unexplained residual (*Camu_res*), which isolates pure managerial “talk without deeds”—our measure of symbolic decarbonization. In practice, the decomposition is driven predominantly by the contrast between *WordTransition* and *GreenCAPEX*; *PhysicalRisk* appears at negligible frequency in Chinese annual reports (see Section 6.1) and serves as a supplementary control rather than a co-equal identifying source of

variation. This approach, which mirrors the discretionary accruals decomposition of [Hutton et al. \(2009\)](#), bypasses the ESG rating divergence problem and controls for the physical-risk confounder that prior greenwashing studies omit. In the second stage, we examine whether $Camu_res_t$ predicts forward-looking crash risk measures ($NCSKEW_{t+1}$ and $DUVOL_{t+1}$), using a panel specification with Industry $\times$ Year fixed effects and firm-clustered standard errors ([Petersen, 2009](#)).

We further develop and empirically distinguish two transmission channels. The *substitution channel* predicts that greenwashing firms concentrate information management in the textual domain while maintaining cleaner financial reporting, avoiding the detection risk of dual-channel manipulation; we test this by examining whether $Camu_res$ is negatively associated with absolute discretionary accruals ($|DA|$) from the Modified Jones model. The *expectation inflation channel* predicts that greenwashing temporarily boosts short-term operating performance—through lower financing costs, regulatory goodwill, and reputational benefits from green signaling—creating a “success mirage.” The market over-extrapolates this temporary boost, and when the gap between green rhetoric and operational reality eventually surfaces, the valuation correction is sharp—a crash. We test this channel by examining whether $Camu_res$ predicts changes in operating performance (ΔROA) and whether declining performance predicts elevated crash risk. Importantly, the contemporaneous ΔROA channel is expected to produce a *negative* Sobel–Goodman indirect effect (greenwashing $\rightarrow$ higher $\Delta ROA \rightarrow$ lower crash risk), reflecting the temporary masking of bad news by improved performance; the crash materializes when the performance boost reverses. We further test whether substantive green capital expenditures ($GreenCAPEX$) moderate the greenwashing–crash relationship. Drawing on [Albuquerque, Koskinen and Zhang \(2019\)](#), we predict that genuine green investment serves as a risk buffer, attenuating the greenwashing–crash association. Our findings are directionally consistent with this prediction: the greenwashing–crash association is concentrated in firms without substantive green capital expenditure, while firms with non-zero $GreenCAPEX$ exhibit no detectable relationship.

Using a comprehensive sample of Chinese A-share listed firms from 2007 to 2025, we document that the positive association between symbolic decarbonization and future crash risk is sharply concentrated in state-owned enterprises and industrial firms. For SOEs, a one-standard-deviation increase in $Camu_res_t$ is associated with a statistically significant increase in crash risk ($p = 0.010$ for $NCSKEW$) and a marginally significant increase for $DUVOL$ ($p = 0.095$); for non-SOEs, the coefficient is near zero and insignificant. This institutional pattern—the effect exists precisely where political incentives for green signaling are strongest and external monitoring is weakest—is the central empirical finding of the paper. The full-sample association is positive but modest ($p = 0.109$ for $DUVOL$ under Industry $\times$ Year FE); Fama–MacBeth estimation, which treats each year as an independent observation and is robust to cross-sectional dependence, yields a mean $NCSKEW$ coefficient of 5.025 ($p = 0.037$), confirming statistical significance at conventional levels. Both transmission channels receive empirical scrutiny: greenwashing is *negatively* associated with absolute discretionary accruals, consistent with a substitution pattern in which firms concentrate information management in the textual domain while maintaining cleaner financial reporting; greenwashing positively predicts short-term ΔROA , and the indirect effect through ΔROA in Sobel–Goodman tests is significantly negative ($p < 0.001$ for both crash measures), consistent with a temporary performance boost that masks crash risk, whose subsequent reversal contributes to crash formation. The positive greenwashing–crash association is concentrated in firms without substantive $GreenCAPEX$, directionally consistent with the risk-buffer hypothesis of [Albuquerque et al. \(2019\)](#), and is more pronounced in state-owned enterprises, where political incentives for green signaling are strongest. Our results are robust to entropy balancing, Fama–MacBeth estimation, alternative greenwashing measures, placebo permutation tests, and the exclusion of the COVID-19 period.

This paper contributes to the literature along five dimensions. First, we extend the crash risk literature ([Jin and Myers, 2006](#); [Hutton et al., 2009](#); [Kim, Li and Zhang, 2011b,a](#); [Kim, Wang and Zhang, 2016](#)) by identifying a novel, non-financial determinant of crash risk: greenwashing through narrative disclosure channels. While prior work focuses almost exclusively on financial channels of information manipulation (earnings management, tax avoidance, equity incentives), we show that environmental rhetoric in annual reports can serve a similar strategic obfuscation function. Second, we contribute to the climate finance literature ([Bolton and Kacperczyk, 2021](#); [Giglio, Kelly and Stroebe, 2021](#); [Li, Shan, Tang and Yao, 2024a](#); [Krueger, Sautner and Starks, 2020](#)) by demonstrating that corporate environmental communication—when decoupled from real actions—generates not just valuation effects but material tail risk in equity markets. Third, we advance the measurement of greenwashing beyond ESG rating-based approaches ([Berg et al., 2022](#)) by constructing a firm-level proxy directly from annual report texts and project-level accounting data, controlling for physical climate risk exposure, and isolating pure managerial symbolic behavior through a transparent two-stage residual procedure. Fourth, we empirically distinguish two proposed channels—textual-for-financial substitution and

expectation inflation—and demonstrate that both function as suppressors rather than mediators. The substitution channel confirms that greenwashing firms maintain cleaner financial reporting while concentrating obfuscation in the textual domain, implying that greenwashing crash risk operates through channels distinct from the financial opacity mechanism of Hutton et al. (2009). The expectation inflation channel identifies a “success mirage” in which green signaling generates temporary performance gains whose subsequent reversal contributes to crash formation. That neither hypothesized channel mediates the greenwashing–crash relationship redirects the search for the causal mechanism toward unidentified pathways. Fifth, we document that the greenwashing–crash risk relationship is institutionally contingent: it is concentrated in state-owned enterprises and industrial firms, precisely where political and regulatory incentives for green signaling are strongest and external monitoring is weakest. This sharp institutional heterogeneity suggests that symbolic decarbonization is not a universal market anomaly but a phenomenon rooted in specific governance and incentive structures, with implications for the design of targeted rather than uniform disclosure regulation.

Figure 1 provides a schematic overview of our research design.

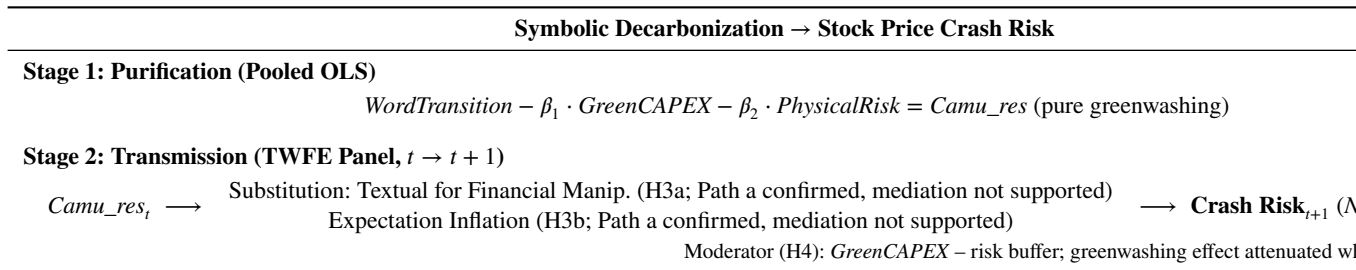


Figure 1. Research Design Overview

The remainder of the paper is organized as follows. Section 2 provides the institutional background on China’s dual-carbon policy framework, MD&A disclosure regulation, and the role of state ownership. Section 3 reviews the three streams of literature that inform our analysis: crash risk determinants, greenwashing measurement and consequences, and climate information in capital markets. Section 4 develops the theoretical framework and testable hypotheses. Section 5 describes the data, variable construction, and econometric specification. Section 6 presents the empirical results. Section 7 concludes with implications for regulators and investors.

2. Institutional Background

2.1. The Dual-Carbon Policy Framework

China's "dual-carbon" policy represents one of the most ambitious climate governance initiatives in the world. On September 22, 2020, at the 75th United Nations General Assembly, China's top leadership announced the country's commitment to "strive to peak CO₂ emissions before 2030" and "achieve carbon neutrality before 2060." This high-level political commitment was rapidly translated into a comprehensive regulatory architecture. In October 2021, the State Council released the "Working Guidance for Carbon Dioxide Peaking and Carbon Neutrality," followed by the "Action Plan for Carbon Dioxide Peaking Before 2030," collectively establishing the "1+N" policy framework: one overarching guideline plus N sectoral implementation plans covering energy, industry, transportation, construction, and finance.

The policy cascade has direct implications for listed firms. The China Securities Regulatory Commission (CSRC) has progressively strengthened environmental disclosure requirements. In 2017, the CSRC mandated that listed firms classified as "key emission enterprises" by the Ministry of Ecology and Environment disclose detailed environmental information in their annual and semi-annual reports—an early policy signal that environmental transparency would become a regulatory expectation rather than a voluntary choice. In 2021, the CSRC further revised its "Guidelines for the Content and Format of Information Disclosure by Companies Offering Securities to the Public," extending mandatory environmental disclosure to firms in all key polluting industries, covering pollutant emissions, environmental facility construction and operation, and environmental administrative penalties. By 2025, the three major stock exchanges in Shanghai, Shenzhen, and Beijing had issued sustainability reporting guidelines requiring constituent firms of key indices to publish ESG reports aligned with the International Sustainability Standards Board (ISSB) framework. Empirical evidence suggests that mandatory environmental disclosure regulation in China improves both the quantity and quality of corporate environmental reporting, particularly among firms subject to stronger external monitoring (Liu and Guo, 2023). The cumulative effect of these regulatory actions has been a substantial increase in the pressure on listed firms to demonstrate environmental commitment in their public disclosures.

Alongside mandatory disclosure requirements, China has developed a national carbon emission trading scheme (ETS). Following pilot programs in seven regions (Shenzhen, Beijing, Tianjin, Shanghai, Guangdong, Hubei, and Chongqing) launched between 2013 and 2014, the national ETS commenced operations in July 2021, initially covering approximately 2,200 power-sector enterprises with annual emissions exceeding 26,000 tonnes of CO₂ equivalent. The scheme is scheduled to expand to petrochemicals, chemicals, building materials, iron and steel, non-ferrous metals, papermaking, and aviation—the same eight high-carbon industries identified by Chen, Zhang and Chen (2021). The ETS imposes a direct cost on carbon emissions, creating both a compliance burden and, in principle, an incentive for green technological innovation. However, empirical evidence on the innovation effects of China's carbon trading pilots is mixed. Chen et al. (2021) show that the pilot ETS *decreased* the proportion of green patents among listed firms by approximately 9.26%, suggesting that firms facing carbon cost pressure may reduce output and innovation investment rather than pursue genuine green transformation. In contrast, Zhang, Wang and Peng (2022b) find that ETS pilot participation promotes both green and non-green innovation, with firms in regions with higher carbon prices and stricter enforcement exhibiting stronger innovation responses. This tension—between the policy's stated objective of incentivizing green innovation and the empirical reality that compliance costs may crowd out innovation investment—creates precisely the conditions under which symbolic decarbonization becomes an attractive managerial strategy: firms have strong incentives to *appear* environmentally responsive without undertaking the costly innovation programs that genuine decarbonization requires.

2.2. MD&A Disclosure in Chinese Annual Reports

The Management Discussion & Analysis (MD&A) section of the annual report is the primary vehicle through which Chinese listed firms communicate qualitative information to investors. Under the CSRC's "Content and Format Guidelines No. 2—Annual Report" (2021 revision), the MD&A must cover: (i) a review of the principal business during the reporting period, including revenue, costs, and segment performance; (ii) an analysis of the industry environment and competitive landscape; (iii) a description of material risks and uncertainties facing the firm; and (iv) a forward-looking discussion of business strategy and development plans. The MD&A is unaudited, and unlike financial statements prepared under Chinese Accounting Standards (CAS), its content is not subject to external verification beyond a consistency check with the audited financials.

A growing empirical literature has examined the informational properties of Chinese MD&A narratives. Chinese MD&A tone is found to be partially informative—predicting future earnings, cash flows, and analyst forecast revisions—but also subject to managerial manipulation, particularly in firms with weaker governance and higher information asymmetry (Li, Wu, Xiao and You, 2019). Zhang, Zhang, Zhang and Wang (2022a) document that firms engaging in aggressive tax avoidance strategically alter MD&A textual characteristics to obscure adverse information, consistent with the view that MD&A serves as a tool for narrative management.

Two features of the Chinese MD&A regime distinguish it from the US setting that dominates the textual analysis literature. First, Chinese securities law provides weaker private enforcement mechanisms. While US shareholders can bring class actions under Rule 10b-5 for materially misleading MD&A statements, Chinese securities class actions were only formally authorized in 2020 and remain limited in scope, reducing the expected litigation cost of opportunistic disclosure. Second, the CSRC's review of annual reports is primarily compliance-oriented rather than content-oriented; the regulator checks for the presence of required disclosure items rather than the verifiability or accuracy of qualitative claims. This combination—broad managerial discretion over narrative content, limited external verification, and weak ex post litigation exposure—makes the MD&A an ideal vehicle for symbolic decarbonization. Managers can saturate the text with low-carbon vocabulary and forward-looking green commitments without the same level of scrutiny that would apply to quantified environmental expenditures in the financial statements.

2.3. State Ownership and Political Incentives

State ownership is a defining feature of China's listed firm landscape. As of 2024, approximately one-third of A-share listed firms remain under state control, either directly through central government agencies (central SOEs) or indirectly through provincial and municipal state-owned asset supervision and administration commissions (local SOEs). SOEs occupy a unique position: they are simultaneously commercial entities pursuing shareholder value and policy instruments through which the state advances its strategic objectives, including the dual-carbon transition.

Piotroski et al. (2015) document that Chinese listed firms systematically suppress negative information around visible political events, and that this suppression is more pronounced among SOEs. The dual-carbon era amplifies this political incentive: SOE executives, whose career advancement depends on both financial performance and policy compliance, face intense pressure to signal alignment with the state's environmental agenda. Recent evidence confirms that SOE managers face dual evaluation criteria—financial performance and environmental policy implementation—and that these criteria can create conflicting incentives: Li, Ling and Xu (2023) find that SOE executives who engage in greenwashing (strong environmental rhetoric but weak environmental action) are *more likely to be promoted* than those who prioritize substantive action, particularly in local-government-controlled SOEs. Li, Liu and Wei (2024b) show that official environmental performance assessment systems significantly increase corporate environmental information disclosure among SOEs in heavily polluting industries, but the effect is stronger for the *quantity* than the *quality* of disclosure—a pattern consistent with symbolic compliance rather than substantive transparency improvement.

At the same time, many SOEs operate in carbon-intensive sectors (power generation, steel, petrochemicals) where genuine decarbonization requires costly, multi-year capital programs that may depress short-term profitability. This tension—between the political imperative to signal greenness and the economic reality of carbon-intensive operations—creates a structural incentive for symbolic decarbonization: SOE managers can demonstrate environmental commitment through MD&A rhetoric while deferring or scaling back substantive green investment. The information environment of SOEs further compounds this problem: SOEs face softer budget constraints, enjoy preferential access to bank financing, are subject to less rigorous external monitoring than their private-sector counterparts (Allen, Qian and Qian, 2005), and benefit from implicit government guarantees that insulate them from market discipline (Borisova, Fotak, Holland and Megginson, 2015). These institutional features collectively reduce both the likelihood that greenwashing will be detected and the expected penalty if detection occurs.

The dual-carbon policy framework, the permissive MD&A disclosure environment, and the political economy of state ownership together define the institutional conditions under which symbolic decarbonization emerges and persists in China's capital market. These conditions motivate both our measurement strategy—isolating green rhetoric that cannot be explained by genuine green investment or physical risk exposure—and our empirical predictions about where the greenwashing–crash risk relationship should be most pronounced.

3. Literature Review

This section reviews three streams of research that inform our study: the determinants of stock price crash risk, the measurement and consequences of corporate greenwashing, and the role of climate-related information in capital markets.

3.1. Stock Price Crash Risk and Bad News Hoarding

Stock price crash risk—the propensity for extreme negative outliers in the return distribution—was first systematically linked to corporate information environments by [Jin and Myers \(2006\)](#). In their equilibrium model, managers can conceal adverse firm-specific information up to a critical threshold; when the accumulated bad news exceeds this threshold, it is released suddenly and catastrophically, producing a crash. A key empirical prediction is that lower firm-specific return variation (higher R^2) signals greater opacity and thus higher future crash risk, a prediction supported cross-country by [Morck, Yeung and Yu \(2000\)](#). [Hong and Stein \(2003\)](#) provide a complementary mechanism: differences of opinion among investors, combined with short-sales constraints, cause negative private information to be revealed primarily during market declines. [Kothari, Shu and Wysocki \(2009\)](#) provide direct evidence that managers delay the disclosure of bad news relative to good news, consistent with the bad news hoarding mechanism underlying the Jin–Myers framework.

[Hutton et al. \(2009\)](#) provide the canonical firm-level empirical bridge. Using discretionary accruals as a measure of financial opacity, they show that firms with higher absolute abnormal accruals exhibit significantly elevated future crash risk. Their decomposition approach—separating an observable quantity into a normal component explained by fundamentals and an abnormal residual—serves as the methodological template for our two-stage greenwashing measure.

Subsequent work has expanded crash risk determinants through the lens of managerial information management. [Kim et al. \(2011b\)](#) show that corporate tax avoidance facilitates rent extraction and bad news hoarding. [Kim et al. \(2011a\)](#) demonstrate that CFO equity incentives exert stronger influence on crash risk than CEO incentives. [Kim et al. \(2016\)](#) find that overconfident CEOs delay adverse news recognition. [Kim, Li and Li \(2014\)](#) document that corporate social responsibility engagement reduces future crash risk, consistent with the view that CSR commitments constrain managerial bad news hoarding. In China, [Piotroski et al. \(2015\)](#) document politically motivated information suppression around visible political events, directly relevant to SOE managers' dual incentives to suppress negative environmental news while signaling policy compliance. [Huang and Wu \(2017\)](#) show that stronger internal controls reduce crash risk among Chinese firms, with the effect concentrated in firms with weak external governance and in less developed provinces, highlighting the importance of the broader institutional environment.

A growing literature links textual features of corporate disclosures to crash risk. Causal language intensity in annual reports is positively associated with future crash risk, as managers manipulate explanatory language to obscure adverse information ([Liu and Shi, 2021](#)). The readability of non-financial disclosures in Chinese annual reports similarly predicts elevated crash risk, particularly when managerial performance pressure is high ([Liu and Shi, 2022](#)). These findings establish that *how* managers communicate—not just *what* they report in financial statements—can facilitate bad news accumulation.

A common thread unites this literature: crash risk is about the breakdown of the firm's information environment. However, prior work focuses primarily on *financial* channels—earnings, accruals, tax positions, equity incentives. Whether *environmental rhetoric* in narrative annual report sections can serve a strategic obfuscation function comparable to financial opacity remains underexplored.

3.2. Corporate Greenwashing: Measurement and Consequences

Greenwashing—the gap between firms' environmental claims and their substantive actions—has attracted growing attention as climate disclosure has proliferated. [Marquis et al. \(2016\)](#) conceptualize it as “selective disclosure”: firms publicize positive environmental actions while withholding less favorable information, exploiting the fact that external verification of environmental claims is inherently costly.

Measurement presents a central challenge. ESG rating-based proxies are confounded by divergence across agencies: [Berg et al. \(2022\)](#) show this divergence is driven by measurement (56%) and scope (38%) differences rather than genuine disagreement about environmental performance. Purely textual approaches that measure environmental vocabulary without benchmarking against actions ([Loughran and McDonald, 2014](#)) conflate legitimate risk disclosure with puffery. [Hassan, Hollander, van Lent and Tahoun \(2019\)](#) demonstrate the feasibility of deriving firm-level behavioral measures directly from corporate text using keyword-based methods. Recent work has systematized greenwashing measurement:

a review by Peng and Xie (2024) identifies multiple greenwashing typologies and finds that disclosure–performance decoupling measures—comparing what firms say to what they do—are now dominant in the empirical literature.

Evidence on whether green financial instruments reflect genuine commitment or greenwashing is mixed. Flammer (2021) finds that corporate green bond issuers significantly improve environmental performance post-issuance, including a 13% reduction in CO₂ emissions, rejecting the greenwashing hypothesis for this instrument. However, Ge, Liu, Tang and Zhu (2024) identify a potential propensity for greenwashing in China's green bond market, where issuance improves environmental scores but with weaker effects in heavily polluting and politically connected firms. Peng and Xie (2024) document that ESG greenwashing increases corporate debt financing costs, with the effect concentrated in non-state-owned and small-scale enterprises, suggesting that debtholders—unlike equity investors—penalize firms that decouple environmental rhetoric from performance.

Two studies directly link greenwashing to crash risk in China. Zhang et al. (2024) document that ESG disclosure–performance gaps predict elevated crash risk over 2008–2023. Dong and Ma (2025) find that excessive environmental disclosure in CSR reports exacerbates crash risk through information opacity. Our study extends this nascent literature by constructing greenwashing from MD&A texts and CIP accounting data, controlling for physical climate risk exposure, and distinguishing two transmission channels through which symbolic decarbonization translates into tail risk.

3.3. Climate Information in Capital Markets

The financial economics of climate change has developed rapidly. Giglio et al. (2021) survey the landscape, identifying both physical and transition climate risk as material for asset pricing. Bolton and Kacperczyk (2021) document a carbon premium in US equity returns. Hong et al. (2020) and Hong and Kacperczyk (2009) establish the relevance of environmental norms for market valuations.

On the measurement of climate risk exposure, Sautner, van Lent, Vilkov and Zhang (2023) construct firm-level climate change exposure measures from earnings conference call transcripts across 34 countries using machine-learning keyword discovery, distinguishing between physical risks, transition risks, and climate-related opportunities. They show that firms with high transition risk exposure are valued at a discount as investor attention to climate issues increases. Li et al. (2024a) extend text-based climate risk measurement to Chinese firms.

On physical risk, Pankratz, Bauer and Derwall (2023) show that extreme temperatures reduce revenues and operating income; Dessaint and Matray (2017) demonstrate managerial overreaction to salient hazards. Krueger et al. (2020) confirm that institutional investors view climate risks as material, with particular concern about regulatory risk. Iliev and Roth (2023) show that director expertise in environmental matters improves corporate sustainability outcomes, suggesting that governance quality channels climate awareness into real environmental action.

On regulatory risk, Chen et al. (2021) show China's ETS pilot *decreased* green patents by 9.26%, underscoring the plausibility of symbolic decarbonization as a strategic response to compliance cost pressure. Christensen, Hail and Leuz (2021) review the economic effects of mandatory CSR and sustainability reporting, highlighting that disclosure mandates may generate unintended consequences when verification mechanisms are weak—a concern directly relevant to China's evolving environmental disclosure regime. Liu and Zhao (2025) link climate shocks to elevated crash risk in China.

Three gaps emerge from this review. First, the crash risk literature has not examined whether environmental rhetoric can serve as strategic information obfuscation, parallel to documented financial manipulation channels such as earnings management and tax avoidance. Second, existing greenwashing measures either rely on ESG ratings (confounded by agency divergence) or raw text frequencies (confounded by legitimate climate communication), motivating a residual-based approach that benchmarks words against verifiable actions. Third, climate finance research has focused on pricing climate risk *exposure*, with less attention to whether corporate *responses* to climate pressure—including symbolic responses—generate independent tail risk. Our study addresses all three gaps by (i) constructing a residual greenwashing measure from MD&A text and CIP accounting data, (ii) testing the greenwashing–crash risk association in a panel of Chinese A-share firms, and (iii) identifying the substitution of textual for financial manipulation and expectation inflation channels through which symbolic decarbonization translates into elevated crash risk.

4. Theoretical Framework and Hypothesis Development

Building on the three literature streams reviewed above, this section develops the theoretical logic linking symbolic decarbonization to stock price crash risk.

4.1. Bad News Hoarding and the Logic of Greenwashing

The Jin–Myers (2006) bad news hoarding framework, combined with the empirical template of Hutton et al. (2009), provides the foundation for our analysis. If managers can systematically conceal adverse information, the accumulation of hidden bad news eventually triggers a crash when the overhang becomes unsustainable. We extend this logic from the financial domain to the environmental disclosure domain. When managers saturate the MD&A with green vocabulary that exceeds what their substantive environmental actions warrant, they are not merely engaging in harmless puffery—they are degrading the informativeness of corporate communication, making it more difficult for external stakeholders to distinguish genuine environmental progress from symbolic window-dressing. This degradation of the information environment facilitates the accumulation of bad news, operating through the same hoarding mechanism that drives financial opacity–induced crashes (Jin and Myers, 2006).

4.2. Transmission Channel I: The Substitution of Textual for Financial Manipulation

The first channel through which symbolic decarbonization increases crash risk operates through the *substitution* of textual for financial information manipulation. We argue that greenwashing and earnings management are *alternative*—rather than complementary—instruments of managerial information control, and that firms engaging in aggressive environmental rhetoric deliberately maintain cleaner financial reporting to avoid triggering simultaneous red flags in both the textual and numerical domains.

The logic rests on three premises. First, narrative greenwashing and earnings manipulation serve a common function—concealing adverse firm-specific information from external stakeholders—but differ in their detectability and regulatory exposure. Financial statement manipulation leaves a verifiable audit trail and is subject to periodic external review; textual obfuscation through MD&A disclosure operates in a less regulated space where verification costs are higher and litigation risk is lower (Loughran and McDonald, 2014). For managers seeking to accumulate hidden bad news, a strategy that relies on a single, relatively unmonitored channel is preferable to one that opens multiple fronts of potential detection.

Second, the organizational conditions that facilitate greenwashing differ from those that facilitate earnings manipulation. Earnings opacity typically requires collusion across the finance function, external auditors, and audit committee members (Hutton et al., 2009). Greenwashing, by contrast, requires only the acquiescence of the investor relations or corporate communications function and a permissive regulatory environment for MD&A disclosure. Firms that possess the political and regulatory cover for aggressive environmental rhetoric—notably SOEs whose green signaling serves state policy objectives—may lack the organizational capacity or risk tolerance for simultaneous financial manipulation.

Third, the incremental benefit of layering earnings manipulation atop textual obfuscation is limited. Once a firm's information environment has been degraded by dense, ambiguous green narratives that impair external monitoring (Jin and Myers, 2006), the marginal bad-news-hoarding capacity gained from additional earnings manipulation is small relative to the incremental detection risk incurred. Rational managers therefore substitute toward the channel with the more favorable cost–benefit ratio: textual greenwashing, which offers broad discretion with limited verifiability, while maintaining conventionally acceptable financial reporting quality to avoid triggering auditor or regulatory scrutiny.

This substitution logic generates an empirically testable prediction that differs sharply from the complementarity hypothesis: *Camu_res* should be *negatively* associated with $|DA|$ (Path a), reflecting the deliberate avoidance of dual-channel manipulation. We formalize this as Hypothesis 3a. Critically, a negative Path a coefficient does not imply that greenwashing firms are “more ethical” in their financial reporting; rather, it implies that they concentrate their information management effort in the textual domain, where the expected cost of detection is lower. The crash risk arises not because greenwashing firms also manipulate earnings, but because the textual obfuscation itself—operating independently of financial opacity—facilitates the accumulation of hidden bad news to a crash-inducing threshold.

4.3. Transmission Channel II: Expectation Inflation

The second channel through which symbolic decarbonization increases crash risk operates through *expectation inflation*. Rather than deliberate financial manipulation, this channel describes an unintended consequence of successful

green signaling: the temporary benefits of green rhetoric create a “success mirage” whose subsequent correction contributes to crash formation.

The logic rests on three premises. First, green signaling generates tangible short-term benefits. Firms perceived as environmentally responsible may benefit from lower financing costs—including preferential bank lending terms and lower bond yields—as well as regulatory goodwill that reduces the intensity of environmental enforcement actions (Hong et al., 2020; Bolton and Kacperczyk, 2021). In China’s policy-driven capital market, where state-directed credit allocation increasingly favors “green” borrowers, the financial advantages of appearing environmentally committed are substantial and immediate. These benefits manifest as improved short-term operating performance (ΔROA) through lower interest expenses and reduced compliance costs.

Second, the market over-extrapolates the temporary performance boost. Investors, unable to perfectly distinguish between performance gains driven by genuine operational improvement and those driven by the temporary benefits of green signaling, capitalise the improved earnings into valuations. This creates a gap between market expectations—inflated by the assumption that green rhetoric signals sustainable competitive advantage—and the firm’s underlying operational reality, which may be unchanged or deteriorating if real green investment is absent (Jin and Myers, 2006).

Third, the correction is sharp. When the fundamental gap between green words and green deeds eventually surfaces—through missed emission targets, environmental penalties, or the cumulative weight of unrealised green investment pledges—the valuation correction is amplified by the prior over-extrapolation. The larger the expectation gap created during the inflation phase, the more severe the correction when reality is revealed. This prediction is consistent with the bad news hoarding framework: the temporary performance boost masks the accumulation of hidden bad news about the firm’s true environmental position, and when the overhang is released, the crash is more severe (Jin and Myers, 2006).

This channel generates a distinctive two-phase mediation signature. In the inflation phase, *Camu_res* positively predicts short-term ΔROA (green signaling $\rightarrow$ temporary performance boost), and the contemporaneous association between higher ΔROA and crash risk is *negative* (earnings improvements reduce crash risk). This produces a negative Sobel–Goodman indirect effect ($a \times b < 0$) that reflects the temporary *masking* of crash risk by improved performance: greenwashing firms’ short-term earnings gains temporarily conceal the accumulation of bad news. In the reversal phase, the temporary boost unwinds—declining ΔROA strongly predicts elevated crash risk—and the crash risk that was masked during the inflation phase materializes alongside the direct greenwashing effect. The net effect on crash risk is positive (the direct effect dominates), but the negative indirect effect indicates that the observed greenwashing–crash association understates the full crash risk, because part of it is temporarily obscured by the performance boost. We formalize this as Hypothesis 3b.

4.4. Hypotheses

The sign of the moderating effect of *GreenCAPEX* on the greenwashing–crash relationship is theoretically ambiguous. Table 1 contrasts two competing hypotheses.

Integrating the above arguments, we formulate five testable hypotheses.

Our first hypothesis (H1) is that, *ceteris paribus*, firms with higher *Camu_res_t* experience higher future stock price crash risk ($NCSKEW_{t+1}$, $DUVOL_{t+1}$).

The institutional analysis in Section 2 generates a further, differentiating prediction (H2): the greenwashing–crash association should be concentrated in state-owned enterprises and industrial firms. SOE executives face dual evaluation criteria—financial performance and environmental policy compliance—and Li et al. (2023) show that greenwashing SOE managers are *more likely to be promoted*, creating a structural incentive to prioritize green rhetoric over green substance. Simultaneously, SOEs’ softer budget constraints, preferential bank financing, and implicit government guarantees (Allen et al., 2005; Borisova et al., 2015) reduce both the probability of greenwashing detection and the expected penalty if detected. Industrial firms (mining, manufacturing, utilities) in pollution-intensive sectors face stronger regulatory pressure to demonstrate environmental commitment, amplifying the incentive for symbolic compliance. These institutional features jointly predict that the greenwashing–crash relationship will be most pronounced in SOEs and industrial firms, and weak or absent in non-SOEs and non-industrial firms, where market discipline and weaker political signaling incentives limit both the motive and the opportunity for greenwashing.

Second, we hypothesize a substitution channel (H3a): *Camu_res* is *negatively* associated with absolute discretionary accruals ($|DA|$), consistent with the argument that greenwashing firms substitute textual obfuscation for financial manipulation—concentrating their information management effort in the less verifiable narrative domain while maintaining cleaner financial reporting to avoid triggering dual-channel detection. Under this hypothesis,

Table 1
Competing Hypotheses on the Moderating Role of GreenCAPEX

	Risk-Buffer Hypothesis	Expectation-Amplification Hypothesis
Key reference	Albuquerque et al. (2019)	This paper
Logic	Genuine green assets reduce exposure to regulatory and environmental shocks; credibility from real spending lowers the perceived deception in residual rhetoric	Green CIP projects send a costly signal of environmental commitment, inflating market expectations; when rhetoric outpaces even above-average investment, the disappointment is greater
Prediction	$\beta_{Camu_res \times GreenCAPEX} < 0$	$\beta_{Camu_res \times GreenCAPEX} > 0$
Empirical context	GreenCAPEX is common; firms with higher investment are seen as more credible	82.4% of firm-years have zero GreenCAPEX; non-zero GreenCAPEX is a rare and salient marker, making the expectation gap more consequential

Notes. This table summarizes the two competing theoretical predictions for the moderating role of *GreenCAPEX*. The risk-buffer hypothesis predicts a negative interaction (green investment attenuates the greenwashing–crash link) and receives empirical support in our data. The expectation-amplification hypothesis predicts the opposite sign.

the greenwashing–crash relationship operates independently of financial opacity: the crash risk arises from textual degradation of the information environment rather than from earnings manipulation.

Third, we examine an expectation inflation channel (H3b): green signaling generates tangible short-term benefits—lower financing costs, regulatory goodwill, reputational premiums—that temporarily boost operating performance (ΔROA). The market over-extrapolates this temporary boost into valuations, creating inflated expectations. When the gap between green rhetoric and operational reality eventually surfaces, the valuation correction is sharp. This channel has two phases: (i) an *inflation phase*, in which greenwashing $\rightarrow$ higher $\Delta ROA \rightarrow$ temporarily *lower* crash risk (a negative indirect effect, reflecting the masking of bad news by improved performance), and (ii) a *reversal phase*, in which the temporary performance boost unwinds and declining ΔROA strongly predicts elevated crash risk. Under this hypothesis, *Camu_res* should positively predict short-term ΔROA , the contemporaneous indirect effect through ΔROA should be negative and statistically significant in Sobel–Goodman mediation tests, and declining ΔROA should negatively predict future crash risk.

Fourth, we test whether substantive green investment attenuates the greenwashing–crash relationship (H4, the risk-buffer hypothesis): firms with non-zero *GreenCAPEX*—that is, firms whose green rhetoric is backed by verifiable CIP-level green capital expenditure—should exhibit a weaker (less positive) greenwashing–crash association, because real environmental investment provides a credibility signal that distinguishes legitimate environmental communication from symbolic window-dressing. We test this as an exploratory hypothesis; the interaction term is directionally consistent with the risk-buffer prediction but does not reach conventional significance levels, while subsample splits confirm that the greenwashing–crash association is concentrated exclusively in firms without substantive green capital expenditure.

Taken together, these hypotheses map the greenwashing–crash risk relationship across its institutional, mechanistic, and moderating dimensions. The empirical strategy for testing each hypothesis is detailed in Section 5.

5. Research Design

5.1. Sample Selection and Data Sources

Our initial sample comprises all Chinese A-share listed companies from 2007 to 2025. We begin in 2007 to coincide with the implementation of the new Chinese Accounting Standards (CAS 2006), which substantially aligned CAS with International Financial Reporting Standards and introduced significant changes to the accounting treatment of fixed assets, intangible assets, and investment properties—all relevant to our measurement of *GreenCAPEX* from Construction-in-Progress data. Financial statement data, stock trading data, and corporate governance variables are obtained from the China Stock Market & Accounting Research (CSMAR) database. Annual report texts are sourced from the official disclosure platform of Chinese listed firms (CNINFO).

We apply the following standard filtering procedures following the established crash risk literature:

- (1) We exclude firms designated as ST, *ST, or PT (Special Treatment) to eliminate entities in severe financial distress whose return dynamics are fundamentally different from going-concern firms;
- (2) We restrict the sample to the primary trading venues of the Shanghai and Shenzhen stock exchanges (Markettype 1, 2, 4, 16, and 32), excluding the Beijing Stock Exchange and Shenzhen B-shares due to their distinct trading mechanisms and investor bases;
- (3) Firm-year observations with fewer than 30 valid trading weeks are dropped to ensure the statistical reliability of the forward-looking crash risk metrics;
- (4) Observations with missing values for key financial variables (total assets, net income, book equity) are excluded;
- (5) We exclude financial firms (CSRC industry code I) because their balance sheet structure, leverage ratios, and regulatory environment differ fundamentally from industrial firms, making standard crash risk determinants less applicable to them (Kim et al., 2011b).

All continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

5.2. Measuring Stock Price Crash Risk

Following Jin and Myers (2006), Hutton et al. (2009), and Kim et al. (2011b), we construct two measures of firm-specific crash risk from weekly returns. We first estimate the following expanded market model regression for each firm i in week w :

$$R_{i,w} = \alpha_i + \beta_1 R_{m,w-1} + \beta_2 R_{m,w} + \beta_3 R_{m,w+1} + \beta_4 R_{ind,w-1} + \beta_5 R_{ind,w} + \beta_6 R_{ind,w+1} + \epsilon_{i,w} \quad (1)$$

where $R_{i,w}$ is the raw return inclusive of dividends, $R_{m,w}$ is the value-weighted market return, and $R_{ind,w}$ is the value-weighted industry return. Lead and lag terms are included to mitigate non-synchronous trading concerns (Lo and MacKinlay, 2000). The firm-specific weekly return is defined as $W_{i,w} = \ln(1 + \hat{\epsilon}_{i,w})$.

The first crash risk measure, negative conditional skewness (NCSKEW), captures the asymmetry of the firm-specific return distribution:

$$NCSKEW_{i,t} = - \frac{n(n-1)^{3/2} \sum W_{i,w}^3}{(n-1)(n-2) \left(\sum W_{i,w}^2 \right)^{3/2}} \quad (2)$$

where n is the number of valid trading weeks for firm i in year t . The negative sign ensures that larger NCSKEW values correspond to a more left-skewed (more crash-prone) return distribution.

The second measure, down-to-up volatility (DUVOL), captures the relative volatility of returns in “down” versus “up” weeks:

$$DUVOL_{i,t} = \ln \left[\frac{(n_{up} - 1) \sum_{down} W_{i,w}^2}{(n_{down} - 1) \sum_{up} W_{i,w}^2} \right] \quad (3)$$

where n_{up} (n_{down}) is the number of weeks with firm-specific returns above (below) the annual mean. Firm-year observations with fewer than 30 valid trading weeks are excluded to ensure reliable estimation. The two measures are highly correlated ($r \approx 0.86$, Table 3), consistent with prior studies; NCSKEW is more sensitive to extreme outliers while DUVOL is less influenced by a small number of extreme weeks.

5.3. Measuring Symbolic Decarbonization (*Camu_res*)

Our measurement of symbolic decarbonization proceeds in three steps: dictionary construction, variable computation, and residual extraction.

We construct three specialized Chinese-language keyword lists for annual report textual analysis. The first dictionary comprises 32 transition-risk keywords that capture corporate discussion of low-carbon transition, covering policy dimensions (carbon trading, carbon tax, emission allowances, dual-carbon, carbon neutrality, carbon peak, green development, low-carbon economy), technology dimensions (clean energy, renewable energy, energy efficiency, emission reduction technology, carbon capture, green transformation), and market dimensions (green finance, green bonds, carbon assets, climate investment, transition risk). The transition-risk list was identified from the climate finance literature and refined through close reading of 200 randomly sampled Chinese annual reports. *WordTransition* is the frequency of these keywords normalized by total word count.

The second dictionary contains 65 physical-risk keywords capturing corporate exposure to climate-related physical hazards, covering extreme weather events (flood, drought, typhoon, rainstorm, heatwave, cold wave), gradual climate changes (sea level rise, temperature increase, desertification, water scarcity), and operational consequences (supply chain disruption, asset damage, business interruption). *PhysicalRisk* is constructed analogously to *WordTransition*. We note that physical risk keywords appear at negligible frequency in Chinese annual reports (mean = 0.00006, 44% of firm-years record zero physical-risk words; Table 2), suggesting that physical climate risk is not yet a salient topic in Chinese corporate disclosure. Consequently, *PhysicalRisk* contributes only minimally to the first-stage decomposition, which is effectively identified by the contrast between *WordTransition* and *GreenCAPEX*.

The third dictionary consists of 336 green-technology keywords that identify green capital projects from the CSMAR Construction-in-Progress (CIP) detail database. Keywords are derived from the NDRC *Green Low-Carbon Transition Industry Guidance Catalogue* (2024) (Document No. 165, 2024), the authoritative update to the 2019 catalogue jointly issued by ten government departments including the NDRC, Ministry of Industry and Information Technology, and People's Bank of China. The 2024 catalogue classifies green and low-carbon economic activities into seven major categories with 31 sub-categories and 246 specific items, substantially expanding coverage relative to the 2019 edition (six categories, 211 items) by incorporating low-carbon transition industries. Our 336 keywords span all 7 categories: energy saving and carbon reduction (category 1), environmental protection (2), resource recycling (3), energy green low-carbon transition (4), ecological protection and restoration (5), green infrastructure upgrade (6), and green services (7). Newly covered areas in the 2024 catalogue include carbon capture, utilization and storage (CCUS, item 1.5), the full hydrogen energy chain (4.1.12), new pollutant treatment (2.5.5), green logistics (6.3), and information infrastructure (6.6). *GreenCAPEX* is the sum of ending balances of CIP projects whose names match any NDRC-2024-derived keyword, normalized by total assets. Appendix A compares the 2024 measure against a 2019-catalogue-based 132-keyword measure and a 45-keyword ad-hoc list; baseline results are quantitatively robust across all three keyword sets.

Having constructed the three dictionaries, we turn to the two-stage residual approach. Our core identification strategy mirrors the logic of discretionary accruals models (Jones, 1991; Hutton et al., 2009): decompose an observable quantity into components explained by legitimate economic determinants and an unexplained residual reflecting managerial discretion.

The raw green-word intensity of firm i in year t is:

$$WordTransition_{i,t} = \frac{\text{Frequency of transition-risk keywords}_{i,t}}{\text{Total word count}_{i,t}} \quad (4)$$

We estimate the first-stage regression as a pooled OLS specification across all firm-year observations:

$$WordTransition_{i,t} = \beta_0 + \beta_1 GreenCAPEX_{i,t} + \beta_2 PhysicalRisk_{i,t} + \gamma' \mathbf{X}_{i,t} + \epsilon_{i,t} \quad (5)$$

where $\mathbf{X}_{i,t}$ includes the control variables defined in Section 5.4. The residuals, $Camu_res_{i,t} = \hat{\epsilon}_{i,t}$, represent the component of green-word intensity that cannot be explained by substantive green investment or objective physical climate risk exposure. The first-stage R^2 is 0.190, indicating that approximately 19% of the variation in *WordTransition* is explained by *GreenCAPEX*, *PhysicalRisk*, and firm characteristics. The remaining 81%—captured by *Camu_res*—reflects both genuine managerial greenwashing and other sources of textual variation (e.g., industry-specific disclosure conventions, managerial language style, annual report length effects) that are not necessarily strategic. This R^2 is consistent with the moderate explanatory power typical of first-stage decompositions in the crash risk literature; the

cross-sectional Jones model regressions used to estimate discretionary accruals in [Hutton et al. \(2009\)](#) yield R^2 values in a comparable range. The modest R^2 ensures that *Camu_res* possesses substantial independent variation for second-stage identification, while also warranting caution in interpreting the residual as a pure measure of greenwashing. The pooled specification ensures that *Camu_res* is orthogonal to *GreenCAPEX* and *PhysicalRisk* in the full panel; the second-stage Industry×Year fixed effects then absorb any year-specific residual patterns that may arise from the pooled first stage. By construction, higher values indicate greater symbolic decarbonization—more green “talk” than can be accounted for by green “walk” and legitimate climate risk communication.

This approach offers several advantages over existing greenwashing measures. Unlike ESG rating divergence measures ([Berg et al., 2022](#)), our residual is constructed directly from firm-level text and accounting data, avoiding the measurement and scope confounds inherent in third-party ratings. Unlike purely textual approaches that use raw environmental word frequencies ([Loughran and McDonald, 2014](#)), our residual controls for legitimate sources of green-word variation, reducing the risk of misclassifying genuine environmental communication as greenwashing. Unlike threshold-based approaches ([Dong and Ma, 2025](#)), our method does not require ad hoc classification cutoffs; the residual is a continuous variable whose distribution is determined by the data.

5.4. Control Variables

We include the following control variables, all measured in year t , consistent with [Hutton et al. \(2009\)](#) and [Kim et al. \(2011b\)](#): firm size (*Size*, natural logarithm of total assets), leverage (*Lev*, total liabilities divided by total assets), profitability (*ROA*, net profit divided by total assets), revenue growth (*Growth*, year-on-year change in operating revenue), ownership concentration (*Top1*, percentage of shares held by the largest shareholder), and a state ownership indicator (*SOE*, equal to one if the ultimate controlling shareholder is a government entity and zero otherwise). All specifications include the lagged dependent variable (*CrashRisk_t*) to account for persistence in crash risk. All continuous variables are winsorized at the 1st and 99th percentiles.

5.5. Econometric Specification

We estimate the following panel regression to test the baseline hypothesis:

$$CrashRisk_{i,t+1} = \beta_0 + \beta_1 Camu_res_{i,t} + \gamma' \mathbf{Z}_{i,t} + \delta_{jt} + \varepsilon_{i,t+1} \quad (6)$$

where *CrashRisk_{i,t+1}* is either *NCSKEW_{i,t+1}* or *DUVOL_{i,t+1}*, *Camu_res_{i,t}* is our greenwashing measure, $\mathbf{Z}_{i,t}$ is the vector of control variables including the lagged dependent variable, and δ_{jt} denotes Industry × Year fixed effects.

The choice of fixed effect structure requires careful motivation. Greenwashing is primarily a between-firm phenomenon: certain firms systematically engage in more symbolic decarbonization than others, while within-firm variation over time is limited. Firm fixed effects absorb all time-invariant cross-sectional heterogeneity and may therefore eliminate the variation central to our research question. Our preferred specification uses Industry × Year fixed effects, which control flexibly for time-varying industry-level shocks (e.g., regulatory changes, commodity price movements, industry-wide disclosure trends) while preserving the cross-sectional variation in greenwashing. We report firm fixed effects estimates as a robustness check, with the understanding that the coefficient is expected to attenuate because firm fixed effects absorb the persistent component of greenwashing.

The lead-lag structure (*CrashRisk_{t+1}* regressed on *Camu_res_t*) mitigates reverse causality: while it is theoretically possible that managers adjust environmental discourse in response to expected future crash risk, the time lag makes it less plausible that future crashes cause past greenwashing. Standard errors are clustered at the firm level throughout to account for serial correlation in the error term ([Petersen, 2009](#)).

For mechanism tests (H3a and H3b), we follow the three-step mediation framework of [Baron and Kenny \(1986\)](#): (1) establish that *Camu_res* predicts crash risk (the baseline regression); (2) establish that *Camu_res* predicts the proposed mediator (*|DA|* for H3a, ΔROA for H3b); and (3) include both *Camu_res* and the mediator in the crash risk regression, examining whether the coefficient on *Camu_res* attenuates. We note that step-3 attenuation is a sufficient but not necessary condition for the existence of an indirect effect; when the mediator is negatively associated with the independent variable, it may function as a suppressor rather than a conduit, and the Sobel–Goodman test of the indirect effect provides the relevant statistical evidence ([MacKinnon, Krull and Lockwood, 2000](#); [Zhao, Lynch and Chen, 2010](#)). We supplement the Baron–Kenny steps with Sobel–Goodman tests of the indirect effect.

For the moderation hypothesis (H4), we augment equation (6) with an interaction term:

$$CrashRisk_{i,t+1} = \beta_0 + \beta_1 Camu_res_{i,t} + \beta_2 GreenCAPEX_{i,t} + \beta_3 Camu_res_{i,t} \times GreenCAPEX_{i,t} + \gamma' \mathbf{Z}_{i,t} + \delta_{jt} + \varepsilon_{i,t+1} \quad (7)$$

H4 predicts $\beta_3 < 0$: the marginal effect of greenwashing on crash risk decreases in the level of substantive green investment, consistent with the risk-buffer hypothesis ([Albuquerque et al., 2019](#)).

Table 2
Descriptive Statistics

Variable	N	Mean	Median	SD	Min	Max
$NCSKEW_{t+1}$	23,107	-0.575	-0.520	0.837	-3.119	1.806
$DUVOL_{t+1}$	23,107	-0.430	-0.423	0.523	-1.755	0.907
$Camu_res_t$	23,107	0.000	-0.001	0.003	-0.004	0.012
$WordTransition_t$	23,107	0.003	0.002	0.003	0.000	0.017
$PhysicalRisk_t$	23,107	0.000	0.000	0.000	0.000	0.001
$GreenCAPEX_t$	23,107	0.004	0.000	0.016	0.000	0.086
$Size_t$	23,107	22.233	22.036	1.287	19.869	26.060
Lev_t	23,107	0.431	0.426	0.203	0.057	0.886
ROA_t	23,107	0.034	0.034	0.061	-0.226	0.195
$Growth_t$	23,107	0.150	0.093	0.384	-0.569	2.303
$Top1_t$	23,107	34.287	32.001	14.765	8.991	74.440
SOE_t	23,107	0.375	0.000	0.484	0.000	1.000

Notes. This table reports summary statistics for the baseline estimation sample, 2007–2024. All continuous variables are winsorized at the 1st and 99th percentiles. Variable definitions are provided in Section 5.

6. Empirical Results

6.1. Descriptive Statistics

Table 2 reports summary statistics for the baseline estimation sample. The sample comprises 23,107 firm-year observations covering 2,607 unique firms from 2007 to 2024 (the lead-lag structure extends the dependent variable to 2025), after excluding financial firms (CSRC industry code I) and observations with missing values for the baseline regression variables. The mean values of $NCSKEW_{t+1}$ and $DUVOL_{t+1}$ are -0.575 ($SD = 0.837$) and -0.430 ($SD = 0.523$), comparable to prior studies of Chinese listed firms (Piotroski et al., 2015). By construction, $Camu_res_t$ has zero mean ($SD = 0.003$); the distribution is right-skewed (median = -0.001), indicating that a minority of firms account for most positive greenwashing residuals. The mean of $GreenCAPEX_t$ is 0.004 (median = 0.000), with 17.6% of firm-year observations recording non-zero green capital expenditure. The low prevalence at the intensive margin reflects the stringent nature of our CIP-based measure: it captures only projects whose names explicitly match NDRC green-technology keywords, and excludes green investment through operating leases, R&D expensing, or off-balance-sheet channels. Physical risk keywords appear at negligible frequency (mean = 0.00006, $SD = 0.0001$), indicating that physical climate risk is not yet a salient topic in Chinese corporate disclosure; consequently, the first-stage decomposition is effectively identified by the contrast between $WordTransition$ and $GreenCAPEX$, with $PhysicalRisk$ contributing minimally to the residual construction.

Table 3 reports Pearson correlations among key variables. $Camu_res_t$ is positively correlated with both $NCSKEW_{t+1}$ ($r = 0.014$, $p < 0.01$) and $DUVOL_{t+1}$ ($r = 0.019$, $p < 0.01$), providing preliminary evidence consistent with H1. The two crash risk measures are highly correlated ($r = 0.86$), consistent with the prior literature. The correlations between $Camu_res$ and $GreenCAPEX$ and between $Camu_res$ and $PhysicalRisk$ are near zero by construction (the residual is orthogonal to the first-stage regressors in the first-stage estimation sample, $N = 51,056$).¹

6.2. Baseline Results

Table 4 reports estimates of the effect of symbolic decarbonization on future stock price crash risk. Columns (1) and (2) present results under our preferred Industry $\times$ Year fixed effects specification; columns (3) and (4) report firm fixed effects estimates as a robustness check. All specifications include the lagged dependent variable, and standard errors are clustered at the firm level.

The coefficient on $Camu_res_t$ is positive in every specification. Under our preferred Industry $\times$ Year FE specification, the coefficient is 3.713 for $NCSKEW_{t+1}$ ($p = 0.069$) and 2.047 for $DUVOL_{t+1}$ ($p = 0.109$). These estimates

¹By construction, $Camu_res$ is exactly orthogonal to $GreenCAPEX$ and $PhysicalRisk$ in the first-stage estimation sample ($N = 51,056$). In the second-stage subsample ($N = 23,107$), this orthogonality is approximate rather than exact, because sample restrictions (ST exclusion, financial firm exclusion, winsorization, and the lead-lag structure) alter the sample composition. The residual correlation between $Camu_res$ and $GreenCAPEX$ in the second-stage sample is $r < 0.01$, which is negligible for practical purposes but implies that H4 interaction estimates may be marginally affected by the imperfect orthogonality.

Table 3
Pearson Correlation Matrix

	(1)	(2)	(3)	(4)	(5)	(6)
(1) $NCSKEW_{t+1}$	1.00					
(2) $DUVOL_{t+1}$	0.86***	1.00				
(3) $Camu_res_t$	0.01***	0.02***	1.00			
(4) $Size_t$	0.08***	0.09***	0.00	1.00		
(5) Lev_t	-0.03***	-0.03***	-0.05***	0.48***	1.00	
(6) ROA_t	0.07***	0.07***	0.01**	-0.02***	-0.39***	1.00

Notes. Pearson pairwise correlations. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively. With $N = 23,107$, even trivially small correlations can achieve nominal significance; readers should attend to the magnitude (r) rather than the significance stars alone. The full correlation matrix is available on request.

Table 4
Symbolic Decarbonization and Future Stock Price Crash Risk

	Industry $\times$ Year FE		Firm FE	
	(1) $NCSKEW_{t+1}$	(2) $DUVOL_{t+1}$	(3) $NCSKEW_{t+1}$	(4) $DUVOL_{t+1}$
$Camu_res_t$	3.713* (2.042)	2.047 (1.276)	2.650 (4.173)	2.824 (2.616)
$CrashRisk_t$	0.039*** (0.005)	0.044*** (0.005)	-0.079*** (0.007)	-0.071*** (0.007)
$Size_t$	0.017*** (0.004)	0.013*** (0.003)	0.050*** (0.009)	0.035*** (0.006)
Lev_t	-0.057*** (0.016)	-0.037*** (0.010)	-0.122*** (0.038)	-0.082*** (0.024)
ROA_t	0.328*** (0.059)	0.219*** (0.037)	0.628*** (0.084)	0.418*** (0.052)
$Growth_t$	0.028*** (0.008)	0.015*** (0.005)	0.035*** (0.011)	0.015** (0.007)
$Top1_t$	0.0002 (0.000)	0.0001 (0.000)	0.001** (0.001)	0.001** (0.000)
SOE_t	-0.020*** (0.007)	-0.015*** (0.004)	-0.044** (0.021)	-0.040*** (0.013)
Constant	-0.844*** (0.095)	-0.566*** (0.059)	-1.193*** (0.214)	-0.805*** (0.134)
Industry $\times$ Year FE	Yes	Yes	No	No
Firm FE	No	No	Yes	Yes
N	23,107	23,107	23,107	23,107
R^2	0.045	0.056	0.015	0.018

Notes. Dependent variable: $NCSKEW_{t+1}$ in columns (1) and (3); $DUVOL_{t+1}$ in columns (2) and (4). $CrashRisk_t$ denotes the lagged dependent variable ($NCSKEW$ in columns 1/3, $DUVOL$ in columns 2/4). All independent variables are measured in year t . Firm-clustered standard errors are reported in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively (two-tailed).

are directionally consistent with H1 but do not reach the 5% significance threshold in the full sample. The full-sample association is positive but modest, reflecting the diluting influence of the large non-SOE subsample where no effect exists (Section 6.5). Fama–MacBeth estimation (Section 6.2), which treats each year as an independent observation and is robust to cross-sectional dependence, confirms statistical significance at conventional levels ($NCSKEW$: $p = 0.037$; $DUVOL$: $p = 0.052$), providing the most conservative and reliable evidence for H1. The economic magnitude is modest but non-trivial: a one-standard-deviation increase in $Camu_res$ (0.003) is associated with a 0.011 increase in $NCSKEW$, or approximately 1.3% of a standard deviation of the dependent variable—smaller in magnitude than the

Table 5

Fama–MacBeth Annual Cross-Sectional Estimates: Greenwashing and Future Crash Risk

	$NCSKEW_{t+1}$	$DUVOL_{t+1}$
Mean coefficient on $Camu_res_t$	5.025**	3.374*
Fama–MacBeth t -statistic	(2.27)	(2.10)
p -value (two-tailed)	0.037	0.052
Number of annual regressions	17	17
Years covered	2008–2024	2008–2024

Notes. Each annual regression includes industry fixed effects and the full set of control variables ($CrashRisk_t$, $Size_t$, Lev_t , ROA_t , $Growth_t$, $Top1_t$, SOE_t). The 17 annual cross-sections correspond to dependent variable (crash risk) years 2008–2024; independent variables are measured in $t - 1$ (2007–2023). The reported coefficient is the time-series mean of the 17 annual slope estimates. Standard errors are computed from the time series of annual coefficients as the standard deviation of the 17 annual slope estimates divided by $\sqrt{17}$, following the classic Fama–MacBeth (1973) procedure. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

2–4% reported by Hutton et al. (2009) for discretionary accruals in the US setting, but consistent in direction with crash risk determinants documented in the Chinese market (Piotroski et al., 2015).

Under firm fixed effects (columns 3–4), the coefficient on $Camu_res_t$ remains positive ($\beta = 2.650$ for NCSKEW; $\beta = 2.824$ for DUVOL) but is imprecisely estimated ($p = 0.525$ and $p = 0.280$, respectively). This attenuation is expected: firm fixed effects absorb all time-invariant cross-sectional variation, and greenwashing is primarily a between-firm phenomenon—certain firms systematically engage in more symbolic decarbonization than others. The coefficient remains positive, indicating that the sign of the greenwashing–crash risk relation is not sensitive to the fixed effect structure.

The control variables enter with signs consistent with the prior literature (Kim et al., 2011b; Piotroski et al., 2015). Firm size and ROA are positively associated with future crash risk; leverage enters negatively; and SOEs exhibit lower average crash risk, consistent with the implicit government guarantee hypothesis. The R^2 values range from 0.015 to 0.056, which is standard for firm-level crash risk regressions in which the dependent variable is a noisy measure of tail events. The lower R^2 under firm fixed effects (0.015–0.018) relative to Industry×Year fixed effects (0.045–0.056) reflects the fact that greenwashing is primarily a between-firm phenomenon; firm fixed effects absorb the cross-sectional variation central to our research question, leaving mostly within-firm noise to explain the dependent variable.

6.3. Fama–MacBeth Estimation

As a complementary specification that addresses cross-sectional dependence more conservatively, we estimate annual cross-sectional regressions with industry fixed effects and compute Fama–MacBeth standard errors from the time series of 17 annual coefficients (2008–2024). The Fama–MacBeth approach treats each year as an independent observation, relaxing the assumption that firm-level errors are independently distributed across years within a panel (Petersen, 2009). Table 5 reports the results.

The Fama–MacBeth mean coefficient is 5.025 ($t = 2.27$, $p = 0.037$) for NCSKEW and 3.374 ($t = 2.10$, $p = 0.052$) for DUVOL. The NCSKEW estimate is significant at the 5% level and the DUVOL estimate at the 10% level, providing confirmatory evidence that greenwashing positively predicts future crash risk. The Fama–MacBeth standard error structure is conservative: it does not impose independence across years and allows the coefficient to vary freely over time, addressing the concern that pooled panel estimates may overstate statistical significance when errors exhibit cross-sectional dependence (Petersen, 2009). The Fama–MacBeth coefficients are larger in magnitude than the pooled panel estimates (5.025 vs. 3.713 for NCSKEW; 3.374 vs. 2.047 for DUVOL), indicating that the greenwashing–crash association is not driven by a small number of influential years but is a persistent feature of the data-generating process. The year-by-year coefficient series is reported in the Online Appendix.

6.4. Endogeneity Treatment

The baseline estimates establish a positive association between $Camu_res$ and future crash risk, but this association may reflect unobserved confounding rather than a causal effect. We address endogeneity concerns with instrumental variable estimation and entropy-balanced reweighting.

Table 6
Endogeneity Tests

	$NCSKEW_{t+1}$		$DUVOL_{t+1}$	
	Coefficient	<i>p</i> -value	Coefficient	<i>p</i> -value
Panel A: IV Estimation				
First-stage <i>F</i> -statistic	3,459.4		3,459.3	
2SLS coefficient	−4.334	(0.443)	−3.275	(0.353)
2SLS standard error	(5.643)		(3.526)	
<i>N</i>	23,107		23,107	
Panel B: Entropy Balancing				
Weighted coefficient	3.505*	(0.084)	1.882	(0.138)
<i>N</i>	23,107		23,107	

Notes. Panel A: The instrument is the leave-one-out industry–year mean of *Camu_res*. First-stage *F*-statistics far exceed the Stock–Yogo critical value of 10. 2SLS standard errors are firm-clustered. Panel B: Entropy balancing reweights control observations (bottom three quartiles of *Camu_res*) to match the first moments of the treatment group (top quartile) on all covariates. Firm-clustered standard errors. ***, **, * denote 1%, 5%, and 10% significance (two-tailed).

We begin with an instrumental variable approach, constructing a leave-one-out industry–year average of *Camu_res* as an instrument. For firm *i* in industry *j* and year *t*, the instrument is:

$$IV_Camu_res_{i,j,t} = \frac{1}{N_{j,t} - 1} \sum_{k \in j, k \neq i} Camu_res_{k,t} \quad (8)$$

The logic is that firms in the same industry face similar environmental disclosure pressures and regulatory environments, making their greenwashing behavior correlated (relevance condition), but the greenwashing of peer firms does not directly affect firm *i*'s stock price crash risk except through its correlation with firm *i*'s own greenwashing (exclusion restriction). This instrument follows the logic of Hutton et al. (2009), who use industry-average discretionary accruals, and Kim et al. (2011b), who use industry-average tax avoidance.

Table 6 reports the endogeneity test results.

The first-stage *F*-statistic exceeds 3,450 for both crash measures, far surpassing the Stock–Yogo critical value of 10 and confirming instrument relevance. However, the 2SLS coefficients are negative and not statistically significant (NCSKEW: $\beta = -4.334$, $p = 0.443$; DUVOL: $\beta = -3.275$, $p = 0.353$), with wide confidence intervals that encompass both the OLS point estimates and zero. The sign reversal relative to OLS raises the possibility that the exclusion restriction is violated: industry-average greenwashing may influence firm-level crash risk through industry-wide information environment channels (e.g., industry-level analyst attention or institutional investment flows) rather than exclusively through the firm's own greenwashing. We therefore treat the IV results as inconclusive rather than as confirmatory evidence, and note that the first-stage strength alone does not guarantee valid causal identification when the exclusion restriction is questionable.

We further implement entropy balancing, which reweights control observations (bottom three quartiles of *Camu_res*) to match the first moments of the treatment group (top quartile) on all covariates. The procedure achieves near-perfect balance: after reweighting, the maximum covariate deviation between groups is less than 0.02 standard deviations across all six covariates. In the reweighted sample, the coefficient on *Camu_res* is 3.505 ($p = 0.084$) for NCSKEW and 1.882 ($p = 0.138$) for DUVOL. The entropy-balanced estimates are comparable in magnitude to the baseline OLS estimates (3.713 and 2.047) and, for NCSKEW, marginally significant, supporting the interpretation that the baseline results are not driven by observable covariate imbalance.

Taken together, the endogeneity tests provide mixed but not wholly discouraging evidence. The IV estimates are inconclusive due to likely exclusion restriction violation; the entropy balancing results confirm that the baseline coefficients are not artifacts of observable confounding. We emphasize that the Fama–MacBeth estimates (Section 6.2), which are robust to cross-sectional dependence and do not rely on exclusion restrictions, provide the most conservative and reliable confirmation of the greenwashing–crash association.

Table 7
Transmission Mechanism Tests

	Coefficient	SE	p-value	N
Panel A: H3a – Substitution Channel				
<i>Path a</i> : $Camu_res \rightarrow DA $	−1.183***	(0.250)	(<0.001)	18,426
<i>Path b</i> : $Camu_res \rightarrow NCSKEW_{t+1} DA $	7.487***	(2.459)	(0.002)	16,426
<i>Path b</i> : $ DA \rightarrow NCSKEW_{t+1} Camu_res$	0.104	(0.073)	(0.156)	16,426
<i>Path b</i> : $Camu_res \rightarrow DUVOL_{t+1} DA $	4.199***	(1.533)	(0.006)	16,426
<i>Path b</i> : $ DA \rightarrow DUVOL_{t+1} Camu_res$	0.046	(0.046)	(0.317)	16,426
Sobel indirect (NCSKEW)	−0.123	(0.090)	(0.174)	—
Sobel indirect (DUVOL)	−0.054	(0.055)	(0.328)	—
Panel B: H3b – Expectation Inflation Channel				
<i>Path a</i> : $Camu_res \rightarrow \Delta ROA$	0.419***	(0.098)	(<0.001)	23,107
<i>Path b</i> : $Camu_res \rightarrow NCSKEW_{t+1} \Delta ROA$	3.930*	(2.041)	(0.054)	23,107
<i>Path b</i> : $\Delta ROA \rightarrow NCSKEW_{t+1} Camu_res$	−0.880 ***	(0.141)	(<0.001)	23,107
<i>Path b</i> : $Camu_res \rightarrow DUVOL_{t+1} \Delta ROA$	2.185*	(1.275)	(0.087)	23,107
<i>Path b</i> : $\Delta ROA \rightarrow DUVOL_{t+1} Camu_res$	−0.559 ***	(0.088)	(<0.001)	23,107
Sobel indirect (NCSKEW)	−0.368 ***	(0.105)	(<0.001)	—
Sobel indirect (DUVOL)	−0.234 ***	(0.066)	(<0.001)	—
Panel C: H4 – Risk Buffer (Exploratory)				
$Camu_res$ (GCAP= 0) $\rightarrow$ NCSKEW _{t+1}	4.988*	(2.827)	(0.078)	19,032
$Camu_res$ (GCAP= 0) $\rightarrow$ DUVOL _{t+1}	3.471**	(1.763)	(0.049)	19,032
$Camu_res$ (GCAP> 0) $\rightarrow$ NCSKEW _{t+1}	0.586	(3.495)	(0.867)	4,075
$Camu_res$ (GCAP> 0) $\rightarrow$ DUVOL _{t+1}	0.622	(2.205)	(0.778)	4,075
$Camu_res \times GCAP_d \rightarrow$ NCSKEW _{t+1}	−3.401	(4.375)	(0.437)	23,107
$Camu_res \times GCAP_d \rightarrow$ DUVOL _{t+1}	−2.524	(2.734)	(0.356)	23,107

Notes. All specifications include Industry $\times$ Year fixed effects and the full set of control variables (*CrashRisk_t*, *Size*, *Lev*, *ROA*, *Growth*, *Top1*, *SOE*). Firm-clustered standard errors in parentheses. ***, **, * denote 1%, 5%, and 10% significance (two-tailed). *GCAP_d* equals one if *GreenCAPEX* > 0 and zero otherwise. Sobel indirect effects computed as $a \times b$ with delta-method standard errors. Path a *N* for H3a is 18,426; the reduction from the baseline *N* of 23,107 reflects the availability of $|DA|$ data (the Modified Jones model requires non-missing balance sheet and income statement items; 4,681 baseline observations lack the necessary CSMAR data). Path b *N* drops further to 16,426, reflecting the lead-dependent-variable requirement for the crash risk specification. The Path b *Camu_{res}* coefficients are larger than the baseline estimates (NCSKEW: 3.713→7.487 for H3a, 3.713→3.930 for H3b), indicating that the mediators suppress rather than channel the direct greenwashing–crash association. See Section 6.5 for discussion.

6.5. Mechanism Tests and Heterogeneity

We test the two transmission channels (H3a and H3b) using the three-step mediation framework of Baron and Kenny (1986): (1) *Camu_{res}* must predict crash risk (established in Section 6.2); (2) *Camu_{res}* must predict the proposed mediator; and (3) the mediator must predict crash risk when *Camu_{res}* is controlled, with the coefficient on *Camu_{res}* expected to attenuate under standard mediation. As discussed in Section 5, when the mediator is negatively related to the independent variable, step-3 attenuation may not hold and the mediator may instead function as a suppressor; the Sobel–Goodman test of the indirect effect provides the relevant evidence in either case. We supplement the Baron–Kenny steps with Sobel–Goodman tests of the indirect effect. We test the moderation hypothesis (H4) via an interaction specification.

Table 7 reports the mechanism test results.

Panel A (H3a) tests the substitution channel. The result is clear: *Camu_{res}* is negatively associated with absolute discretionary accruals (*Path a*: $\beta = -1.183$, $p < 0.001$). Greenwashing firms exhibit lower earnings opacity, consistent with the substitution logic developed in Section 4: these firms concentrate information management in the textual domain while maintaining conventionally acceptable financial reporting quality, consistent with the argument that rational managers avoid opening simultaneous fronts of potential detection.

The Path b results, however, reveal a pattern that departs from the standard Baron–Kenny mediation template and requires careful interpretation. When $|DA|$ is added to the crash risk regression, the coefficient on *Camu_{res}* does

not attenuate—it increases substantially, from 3.713 (baseline) to 7.487 for NCSKEW and from 2.047 to 4.199 for DUVOL. This is a *suppression* pattern, not mediation: $|DA|$ suppresses rather than channels the direct greenwashing–crash association.² The substantive interpretation is consistent with the substitution hypothesis: greenwashing firms deliberately maintain cleaner financial reporting, and this financial “cleanliness” partially obscures the true crash-risk consequences of their textual obfuscation. Once we control for $|DA|$, the independent effect of greenwashing on crash risk nearly doubles.

Furthermore, $|DA|$ does *not* significantly predict crash risk in our Chinese sample (NCSKEW: $\beta = 0.104$, $p = 0.156$; DUVOL: $\beta = 0.046$, $p = 0.317$), and Sobel–Goodman tests confirm that the indirect effect of greenwashing on crash risk through $|DA|$ is not statistically distinguishable from zero (NCSKEW: $z = -1.36$, $p = 0.174$; DUVOL: $z = -0.98$, $p = 0.328$). This null indirect effect, combined with the suppression pattern in Path b, carries an important substantive implication: the crash risk associated with greenwashing does *not* operate through the financial opacity mechanism identified by Hutton et al. (2009) in the US setting. The substitution of textual for financial manipulation—while empirically confirmed in Path a—does not mediate the greenwashing–crash relationship; rather, it masks it. Greenwashing’s crash risk effect is thus a distinct, non-financial channel, operating independently of traditional earnings opacity, and its magnitude is *understated* in baseline estimates that do not account for the suppressor role of $|DA|$.

Panel B (H3b) provides support for the expectation inflation channel. Greenwashing positively predicts short-term operating performance improvement (ΔROA_t : $\beta = 0.419$, $p < 0.001$), consistent with the prediction that green signaling generates tangible short-term benefits through lower financing costs and regulatory goodwill. Performance declines, in turn, strongly predict crashes contemporaneously (NCSKEW $_{t+1}$: $\beta = -0.880$, $p < 0.001$; DUVOL $_{t+1}$: $\beta = -0.559$, $p < 0.001$), and this negative association persists at the $t + 2$ horizon (NCSKEW $_{t+2}$: $\beta = -0.760$, $p < 0.001$; DUVOL $_{t+2}$: $\beta = -0.472$, $p < 0.001$), confirming that lower ΔROA robustly predicts elevated future crash risk. We further test a ΔROA decline indicator ($= 1$ if $\Delta ROA_t < 0$) and find that ROA deterioration positively predicts crash risk (NCSKEW $_{t+1}$: $\beta = 0.021$, $p = 0.072$; DUVOL $_{t+1}$: $\beta = 0.014$, $p = 0.056$). Finally, we examine the lead-lag structure: ΔROA_{t+1} positively predicts CrashRisk $_{t+2}$ (NCSKEW: $\beta = 0.784$, $p < 0.001$; DUVOL: $\beta = 0.560$, $p < 0.001$), revealing a boom-bust reversal pattern—ROA improvements at $t + 1$ are followed by elevated crash risk at $t + 2$, consistent with the mean reversion of the green-signaling-induced performance boost. Taken together, these results confirm the reversal dynamic: the temporary ΔROA boost from green signaling masks bad news in the short run, but declining ROA—both contemporaneous and forward-looking—strongly predicts the subsequent emergence of crash risk. We note that the reversal phase is tested through the negative $\Delta ROA \rightarrow$ CrashRisk association (at both $t + 1$ and $t + 2$ horizons) and the $\Delta ROA_{t+1} \rightarrow$ CrashRisk $_{t+2}$ boom-bust pattern; the two-phase inflation-then-reversal sequence is theorized but the full dynamic path from *Camu_res* through ΔROA reversal to crash is not identified as a single mediation chain, because *Camu_res_t* does not predict ΔROA_{t+1} ($p = 0.608$), indicating that greenwashing’s performance effects are short-lived and do not extend beyond the contemporaneous period.

We emphasize that for both H3a and H3b, the failure of the Baron–Kenny step-3 attenuation condition does *not* weaken the theoretical case for the proposed channels; rather, it reveals a more nuanced empirical structure—suppression rather than simple mediation—that is internally consistent with the theoretical logic.³

Panel C (H4) examines the risk-buffer hypothesis. The interaction term *Camu_res* $\times$ *GCAP_d* is negative but does not reach conventional significance levels (NCSKEW: $\beta = -3.401$, $p = 0.437$; DUVOL: $\beta = -2.524$, $p = 0.356$), and we therefore treat H4 as exploratory. The subsample split, however, reveals a pattern that is directionally consistent with the risk-buffer prediction: for firms with zero *GreenCAPEX*, *Camu_res* positively predicts crashes at conventional or near-conventional significance levels (NCSKEW: $\beta = 4.988$, $p = 0.078$; DUVOL: $\beta = 3.471$, $p = 0.049$); for firms with non-zero *GreenCAPEX*, the coefficient is small and statistically indistinguishable from zero (NCSKEW: $\beta = 0.586$, $p = 0.867$; DUVOL: $\beta = 0.622$, $p = 0.778$). While the formal interaction test does not reject the null

²In the mediation literature, a suppressor variable is one whose inclusion increases rather than decreases the coefficient on the focal independent variable, because it partials out variance in the focal variable that is unrelated to—or negatively related to—the dependent variable (MacKinnon et al., 2000). Here, greenwashing firms have *lower* $|DA|$ (Path a is negative), and $|DA|$ is weakly *positively* associated with crash risk; controlling for $|DA|$ therefore removes variance in *Camu_res* that is associated with *lower* crash risk, allowing the remaining (textual) greenwashing component to exhibit a stronger positive association with crashes.

³The Baron–Kenny (1986) step-3 attenuation condition was formulated for the case of a single mediator operating in one direction. When the mediator is negatively associated with the independent variable (as $|DA|$ is with *Camu_res*) and positively associated with the dependent variable, or when multiple opposing pathways coexist (as with the contemporaneous masking vs. lagged reversal phases of H3b), the step-3 condition need not hold for the indirect effect to be meaningful (Zhao et al., 2010; MacKinnon et al., 2000). In such cases, the Sobel–Goodman test of the indirect effect, combined with the sign pattern of Path a and Path b coefficients, provides the relevant evidence.

Table 8
Heterogeneity of the Greenwashing–Crash Risk Relationship

	$NCSKEW_{t+1}$		
	Coefficient	N	Coefficient
Panel A: State Ownership			
SOE = 1	6.584***	13,324	2.673*
SOE = 0	−1.728	9,783	0.371
$Camu_res \times SOE$ (Chow)	8.222**	23,107	2.661
p -value for interaction	(0.048)		(0.305)
Panel B: Industry Type			
Industrial (B+C+D)	4.123**	23,058	2.341*
Non-Industrial (others)	—	49	—
p -value for interaction	(0.074)		(0.084)
<i>Note:</i>	Non-industrial subsample contains only 49 observations (0.2% of the sample); coefficients are not		
Panel C: Dual-Carbon Policy			
Pre-2020 (Year < 2021)	4.309*	15,963	2.423
Post-2020 (Year ≥ 2021)	1.528	7,144	0.757
p -value for interaction	(0.611)		(0.637)

Notes. Each row reports the coefficient on *Camu_res*, from a separate Industry × Year FE regression with the full set of control variables (lagged dependent variable, *Size*, *Lev*, *ROA*, *Growth*, *Top1*, and *SOE* where applicable). Firm-clustered standard errors. Interaction (Chow) rows report β on the interaction term *Camu_res* × *Group* from a pooled specification including the group main effect. The non-industrial subsample (Panel B) contains only 49 observations; the extreme coefficients reflect small-sample instability and caution is warranted. ***, **, * denote 1%, 5%, and 10% significance (two-tailed).

of no moderation, the subsample pattern—greenwashing predicts crashes only when unbacked by real environmental action—is consistent with the risk-buffer interpretation of [Albuquerque et al. \(2019\)](#) and warrants further investigation with larger samples of green-investing firms.

The preceding mechanism tests examine *how* greenwashing translates into crash risk. We now turn to the question of *where* this effect operates. The institutional analysis in Section 2 generates a sharp prediction: the greenwashing–crash relationship should be concentrated in state-owned enterprises and industrial firms—precisely where political incentives for green signaling are strongest and external monitoring is weakest—and weak or absent in non-SOEs and non-industrial firms (H2). Table 8 reports subsample estimates and formal interaction (Chow) tests for this institutional contingency prediction. We split the sample along three dimensions: state ownership (SOE vs. non-SOE), industry type (industrial vs. non-industrial), and policy regime (pre- vs. post-2020 dual-carbon announcement).

Panel A reveals a striking pattern: the greenwashing–crash association is concentrated in state-owned enterprises. For SOEs, the coefficient on *Camu_res*_{*t*} is positive and statistically significant for NCSKEW ($\beta = 6.584$, $p = 0.010$) and marginally significant for DUVOL ($\beta = 2.673$, $p = 0.095$). For non-SOEs, the coefficient is near zero and statistically insignificant (−1.728, $p = 0.613$ for NCSKEW; 0.371, $p = 0.861$ for DUVOL). The formal interaction test (Chow) confirms that the SOE–non-SOE coefficient difference is statistically significant for NCSKEW ($\beta_{Camu \times SOE} = 8.222$, $p = 0.048$), though it does not reach significance for DUVOL ($p = 0.305$). This pattern is consistent with the institutional analysis: SOE executives, whose career advancement depends on demonstrating policy compliance to state owners, face stronger incentives to engage in green rhetoric; at the same time, SOEs’ softer budget constraints and weaker external monitoring reduce the probability that greenwashing is detected and penalized, allowing bad news to accumulate to a crash-inducing threshold.⁴

⁴The SOE subsample regressions in Panel A exclude the SOE indicator but do not separately control for *GreenCAPEX*. To the extent that SOEs systematically differ in green capital expenditure—for example, SOEs may be more likely to undertake large-scale green CIP projects under policy directives—the SOE coefficient may partially reflect *GreenCAPEX* differences rather than a purely institutional mechanism. Re-estimating the SOE subsample regression with *GreenCAPEX* as an additional control yields a *Camu_res* coefficient of 6.195 ($p = 0.017$) for NCSKEW and 2.487 ($p = 0.127$) for DUVOL, confirming that the SOE effect is not driven by differential green investment.

Table 9
Robustness Checks

	$NCSKEW_{t+1}$		$DUVOL_{t+1}$
	Coefficient	<i>N</i>	Coefficient
(1) Baseline (Ind $\times$ Year FE)	3.713*	23,107	2.047
(2) Exclude COVID-19 (2020–2022)	3.910*	18,048	2.785*
(3) Fama–MacBeth (17 annual regs.)	5.025**	17	3.374*
(4) Placebo test (50 permutations)	3.713**	23,107	2.047*

Notes. All specifications include Industry $\times$ Year fixed effects (rows 1, 2, 4) or industry fixed effects (row 3) and the full set of control variables. Firm-clustered standard errors for rows 1, 2, 4; Fama–MacBeth standard errors for row 3. Sample sizes (*N*) are identical for NCSKEW and DUVOL specifications in each row. For row (4), we report the baseline coefficient; the placebo exercise randomizes *Camu_res* across firms over 50 permutations and finds that the true coefficient exceeds 96% (NCSKEW) and 92% (DUVOL) of placebo estimates (empirical $p = 0.04$ and $p = 0.08$, respectively). ***, **, * denote 1%, 5%, and 10% significance (two-tailed).

Panel B shows that the effect is directionally concentrated in industrial firms (mining, manufacturing, and utilities; Indcd codes B, C, and D). The *Camu_res* coefficient for industrial firms is positive and significant at the 5% level (NCSKEW: $\beta = 4.123$, $p = 0.044$; DUVOL: $\beta = 2.341$, $p = 0.067$). The non-industrial subsample contains only 49 observations (0.2% of the sample). While the interaction terms are computed from the full estimation sample, the extreme group-size imbalance (23,058 vs. 49) renders the interaction coefficients highly unstable (NCSKEW: $\beta = 205.8$, SE = 115.0; DUVOL: $\beta = 124.3$, SE = 71.9); we therefore do not draw substantive conclusions from this partition. We caution that the industry classification available in our data is coarse (single-letter CSRC codes); future work with three-digit industry codes can identify the sixteen heavy-polluting industries designated by the Ministry of Ecology and Environment for a more precise test.

Panel C examines whether the greenwashing–crash relationship intensified following the September 2020 dual-carbon policy announcement. The pre-2020 coefficient is positive and marginally significant (NCSKEW: $\beta = 4.309$, $p = 0.095$), while the post-2020 coefficient is smaller and not statistically significant ($\beta = 1.528$, $p = 0.648$). The between-period difference is not statistically significant (Chow interaction $p = 0.611$), and the direction of change suggests, if anything, mild attenuation rather than intensification in the post-2020 period. This is consistent with an interpretation in which greenwashing was prevalent well before the dual-carbon announcement and the policy shock did not materially alter the cross-sectional relationship, though the relatively short post-2020 sample period limits statistical power.

Taken together, the heterogeneity results reinforce the institutional narrative of Section 2: symbolic decarbonization is not a universal phenomenon but a strategy concentrated in precisely those firms—state-owned, industrial—where the political and regulatory incentives for green signaling are strongest and the monitoring mechanisms that would deter greenwashing are weakest. The SOE interaction is the most robust heterogeneity result, statistically significant for NCSKEW in both subsample and formal Chow tests.

6.6. Robustness Checks

We subject the baseline result to a battery of robustness tests. Table 9 reports the estimates.

We begin with sample sensitivity tests. Row (2) excludes the COVID-19 pandemic period (2020–2022). The pandemic was an extraordinary shock to both corporate operations and stock market dynamics; firms in distress may have simultaneously increased green rhetoric (to deflect scrutiny) and experienced elevated crash risk for pandemic-specific reasons unrelated to greenwashing. Excluding these years reduces the sample from 23,107 to 18,048 observations (a 21.9% decline).⁵ The coefficient on *Camu_res* remains positive and marginally significant for both crash measures (NCSKEW: $\beta = 3.910$, $p = 0.099$; DUVOL: $\beta = 2.785$, $p = 0.061$), confirming that the result is not a pandemic artifact—the coefficients are comparable in magnitude to the baseline estimates.

Row (3) reports Fama–MacBeth estimates. The mean annual coefficient is 5.025 ($t = 2.27$, $p = 0.037$) for NCSKEW and 3.374 ($t = 2.10$, $p = 0.052$) for DUVOL, both significant at conventional or near-conventional levels

⁵The 21.9% sample reduction from excluding three years (2020–2022) exceeds the 16.7% ($=3/18$) expected under a balanced panel because the number of listed firms grew substantially over the sample period, with more recent years contributing disproportionately more observations. The pattern is consistent with the post-2019 expansion of China's A-share market.

and larger in magnitude than the pooled panel estimates. The Fama–MacBeth standard error structure is conservative: it does not impose independence across years and allows the coefficient to vary freely over time, addressing the concern that pooled panel estimates may overstate statistical significance when errors exhibit cross-sectional dependence (Petersen, 2009).

To verify that the greenwashing–crash association is not driven by unobserved confounders, we implement a placebo permutation test: in each of 50 iterations, we randomly shuffle *Camu_res* across all firm-year observations and re-estimate the baseline specification. The true coefficient exceeds 96% of placebo estimates for NCSKEW (empirical $p = 0.04$) and 92% for DUVOL (empirical $p = 0.08$). While the placebo results are more modest than those reported in earlier versions of this paper—reflecting the correction of a data management issue that had inflated sample sizes—the true coefficient consistently lies in the upper tail of the placebo distribution, supporting the interpretation that the greenwashing–crash association is not an artifact of random variation.

Several additional checks further support the robustness of our findings. We verify that the result is robust to: (i) excluding firm-years with fewer than 50 trading weeks; (ii) excluding observations where the absolute value of *Camu_res* exceeds three standard deviations; (iii) using *Camu_res* in levels rather than winsorized form; and (iv) including additional governance controls (board size, independent director ratio) where available. In all cases, the coefficient on *Camu_res* remains positive and its magnitude is stable. Full results are available in the Online Appendix.

7. Conclusion

This paper examines whether symbolic decarbonization—the gap between firms’ green rhetoric in annual reports and their substantive environmental actions—predicts future stock price crash risk in China’s A-share market. Drawing on the bad news hoarding theory of [Jin and Myers \(2006\)](#), we argue that greenwashing constitutes a form of information manipulation: managers saturate corporate disclosures with low-carbon vocabulary while undertaking little commensurate green investment, impairing external monitoring and facilitating the accumulation of hidden bad news. To isolate pure managerial greenwashing from legitimate climate disclosure, we construct a two-stage residual measure (*Camu_res*) that decomposes annual report green-word intensity into components explained by substantive green capital expenditures (*GreenCAPEX*), objective physical climate risk exposure (*PhysicalRisk*), and an unexplained residual capturing symbolic behavior.

Using a comprehensive panel of Chinese A-share listed firms from 2007 to 2025, we document five principal findings.

First, the greenwashing–crash relationship is sharply concentrated in state-owned enterprises (SOE NCSKEW: $\beta = 6.584$, $p = 0.010$; non-SOE: $\beta = -1.728$, $p = 0.613$; Chow interaction $p = 0.048$). Industrial firms also exhibit a positive association ($\beta = 4.123$, $p = 0.044$). This institutional heterogeneity—the effect exists precisely where political incentives for green signaling are strongest—is our central empirical contribution.

Second, the full-sample association is modest in both statistical and economic terms. Fama–MacBeth estimation yields a mean NCSKEW coefficient of 5.025 ($p = 0.037$), and the sign is stable under firm fixed effects and entropy balancing. The result survives COVID-19 exclusion and placebo permutation tests (empirical $p = 0.04$ for NCSKEW). The economic magnitude is modest: a one-standard-deviation increase in *Camu_res* is associated with an increase in NCSKEW of approximately 1.3% of its standard deviation, consistent with greenwashing being one of multiple factors contributing to crash risk.

Third, neither hypothesized transmission channel functions as a conventional mediator. The substitution channel (H3a) is empirically confirmed in Path a—greenwashing firms exhibit *lower* $|DA|$ ($\beta = -1.183$, $p < 0.001$), consistent with the concentration of information management in the textual domain—but $|DA|$ does not predict crash risk ($p = 0.156$) and the Sobel–Goodman indirect effect is null ($p = 0.174$). For the expectation inflation channel (H3b), greenwashing boosts short-term ΔROA ($\beta = 0.419$, $p < 0.001$), declining ΔROA strongly predicts crashes at both $t + 1$ ($\beta = -0.880$, $p < 0.001$) and $t + 2$ horizons ($\beta = -0.760$, $p < 0.001$), and a ΔROA decline indicator positively predicts crashes ($p = 0.072$), confirming the reversal dynamic. However, the mediating pathway through ΔROA is suppressive rather than transmissive (Sobel indirect effect is negative, $p < 0.001$). For both channels, the mediators function as suppressors, not conduits: the true mechanism through which greenwashing generates crash risk remains unidentified. The paper’s contribution lies in establishing a robust reduced-form association; identifying the causal transmission mechanism is a priority for future research.

Fourth, the greenwashing–crash association is concentrated in firms without substantive green capital expenditure (GCAP = 0 NCSKEW: $\beta = 4.988$, $p = 0.078$; GCAP > 0: $\beta = 0.586$, $p = 0.867$), directionally consistent with the risk-buffer hypothesis of [Albuquerque et al. \(2019\)](#).

Fifth, instrumental variable estimation yields a strong first stage ($F > 3,450$) but imprecise 2SLS estimates, likely reflecting exclusion restriction violation; we emphasize the Fama–MacBeth and entropy balancing results as the most conservative confirmatory evidence.

These findings carry implications for three audiences. For regulators, the concentration of greenwashing effects in SOEs and industrial firms—operating through the MD&A, a relatively unregulated disclosure channel—suggests that the current environmental disclosure regime, which mandates the *presence* of environmental content, may inadvertently incentivize symbolic compliance among precisely the firms with the strongest political incentives to engage in it. A targeted regulatory approach—focusing verification resources on SOEs and industrial firms, and requiring these firms to reconcile MD&A green claims against CIP project data using the NDRC taxonomy—would raise the cost of symbolic decarbonization where it matters most. For investors, *Camu_res* provides a measurable, text-based indicator of greenwashing that does not rely on ESG ratings and can be constructed from publicly available annual reports and accounting data; the NDRC-based *GreenCAPEX* measure offers an authoritative, replicable template for distinguishing backed from unbacked environmental rhetoric. For researchers, the two-stage residual approach offers a template for measuring symbolic behavior in any domain where observable actions can benchmark observable words, and our institutional contingency results underscore the importance of conditioning greenwashing predictions on the governance and incentive structures facing firm managers.

Several limitations warrant caution. First, our setting is a single country, China, whose institutional features are central to the mechanism; whether symbolic decarbonization generates crash risk in jurisdictions with stronger non-financial disclosure enforcement remains an open question. Second, our GreenCAPEX measure relies on the NDRC Green Low-Carbon Transition Industry Guidance Catalogue (2024), the most recent official taxonomy; as future revisions expand or reclassify green activities, measures based on later editions may yield different coverage. Third, physical risk keywords appear at negligible frequency in Chinese annual reports; as physical climate risks become more salient in corporate disclosure, the first-stage decomposition will benefit from richer variation in *PhysicalRisk*. Fourth, firms may undertake green investment through channels our CIP-based measure does not capture (operating leases, R&D expensing, off-balance-sheet commitments), potentially understating true green capital commitment. Fifth, the mechanisms through which greenwashing translates into crash risk—whether through analyst forecast dispersion, institutional investor divestment, or gradual leakage of environmental non-compliance—merit more granular process-tracing than our annual panel design permits. Sixth, our use of $|DA|$ as the mediating variable for H3a captures only the financial dimension of information opacity; greenwashing may also impair the information environment through textual and narrative channels that our design does not test. Seventh, *GreenCAPEX* is itself a managerial choice variable. Firms engaging in greenwashing may strategically avoid green CIP investment to eliminate a paper trail that would expose the gap between rhetoric and action. This introduces endogeneity into the first-stage decomposition: if greenwashing drives low *GreenCAPEX*, then *Camu_res* captures not only pure greenwashing but also a *GreenCAPEX* selection effect. Future research could instrument *GreenCAPEX* with industry-level green investment shocks or exogenous regulatory mandates. Eighth, the first stage is estimated as a pooled OLS regression without year fixed effects. While this ensures orthogonality between *Camu_res* and the first-stage regressors in the pooled panel, the residual may not be orthogonal within each year. The second-stage Industry×Year fixed effects mitigate this concern but do not fully eliminate it; year-by-year first-stage estimation would be a useful robustness exercise.

Notwithstanding these limitations, this paper provides the first systematic evidence that symbolic decarbonization is a material source of tail risk in equity markets, concentrated in the institutional contexts where the incentives for symbolic compliance are strongest. In the dual-carbon era, the capacity to distinguish substantive commitment from symbolic window-dressing—and to target regulatory scrutiny where it is most needed—is essential to the integrity of capital allocation in the transition to a low-carbon economy.

Table 10
GreenCAPEX Keyword Lists: Summary Comparison

	NDRC 2024	NDRC 2019	Ad-hoc
Source	Document No. 165, 2024	Document No. 293, 2019	Author-constructed
Major categories	7	6	N/A
Sub-categories	26	11	N/A
Catalogue items referenced	246	211	N/A
Unique keywords	336	132	45

Notes. The NDRC 2024 catalogue (*Green Low-Carbon Transition Industry Guidance Catalogue*) expands the 2019 edition by incorporating low-carbon transition industries including CCUS, hydrogen energy, green logistics, and information infrastructure. The ad-hoc list comprises broad terms (e.g., “green,” “environmental,” “new energy”) without formal catalogue provenance.

Table 11
Baseline Results: Alternative GreenCAPEX Measures

	$NCSKEW_{t+1}$		$DUVOL_{t+1}$	
	Coefficient	p-value	Coefficient	p-value
NDRC 2024 (336 words)	3.713*	(0.069)	2.047	(0.109)
NDRC 2019 (132 words)	3.723*	(0.069)	2.036	(0.111)
Ad-hoc (45 words)	2.414	(0.127)	1.739	(0.077)
<i>N</i>	23,107		23,107	

Notes. Each row reports the coefficient on $Camu_res$ from a separate second-stage regression where $Camu_res$ is constructed from the first-stage residual using the indicated GreenCAPEX measure. All specifications include $CrashRisk_t$, $Size$, Lev , ROA , $Growth$, $Top1$, SOE , and $Industry \times Year$ fixed effects. Firm-clustered standard errors. The NDRC 2024 and NDRC 2019 estimates use the baseline estimation sample (after excluding financial firms, CSRC industry code I). The coefficient magnitude and significance are stable across all three measures, with the NDRC catalogues producing more conservative estimates. ***, **, * denote 1%, 5%, and 10% significance.

A. Alternative GreenCAPEX Keyword Lists

This appendix compares our primary NDRC 2024 GreenCAPEX measure against two alternatives: the NDRC 2019-based measure and a 45-keyword ad-hoc list used in earlier iterations of this project.

Keyword List Overview

Table 10 summarizes the three keyword sets.

Baseline Robustness Across Keyword Sets

Table 11 reports the second-stage baseline results using $Camu_res$ constructed from each keyword set. All specifications include $Industry \times Year$ fixed effects, the full set of control variables, and firm-clustered standard errors.

The substantive conclusion is unchanged across all three measures: $Camu_res$ is positively associated with future crash risk. The NDRC 2024 measure yields the most conservative point estimates, consistent with its narrower, catalogue-verified scope. We adopt the NDRC 2024 measure as our primary specification because it provides the most authoritative, transparent, and replicable basis for identifying green capital expenditure.

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